

Deep Learning Final Project: BirdCLEF+ 2026

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Abstract

BirdCLEF+ 2026 requires classifying 234 Pantanal species from noisy field audio, scored by macro ROC-AUC and evaluated offline on Kaggle CPUs under a 90-minute budget. We build a two-stage pipeline—frozen Google Perch-v2 embeddings (1,536-D) followed by a lightweight Mixture-of-Experts (MoE) head exported to ONNX—and optimize it through a series of experiments, reporting what worked and what did not. The largest single gain came from 50% overlapping test-time augmentation (TTA, +0.025 public AUC); Focal Loss replaced an underperforming binary cross-entropy baseline; a late addition of BatchNorm stabilized routing; and 5-fold ensembling lifted a single fold from 0.742 to 0.839 private. Several promising ideas *failed*: embedding mixup and waveform background mixing both over-regularized into “alien” samples, a top-1 sparse MoE collapsed onto a single expert, larger temporal context overfit site acoustics, geographic masking did not help, and an iterative pseudo-labeling (Noisy-Student) loop was abandoned due to marginal gains and excessive complexity. Data analysis exposed the project’s recurring obstacles—severe class imbalance, multiple data-leakage risks, and a focal→soundscape domain shift that decouples local validation from the leader board. Our final 5-fold ONNX ensemble scores **0.839** private macro ROC-AUC.

1 Introduction

BirdCLEF+ 2026 asks participants to identify 234 scored Pantanal species—birds, amphibians, insects, mammals and a single reptile—from 5-second windows of passive acoustic monitoring (PAM) audio. The metric is **macro averaged ROC-AUC**, which averages a per-class ranking score with equal weight on every class and skips classes absent from the hidden test. Inference runs in a **network-disabled Kaggle notebook** under a **90-minute CPU budget**, ruling out heavy end-to-end CNNs and pushing our design considerations toward precomputed features, heavy data pre-processing, and an efficient runtime.

Our objective was to build a deployable pipeline and improve it empirically, aiming for a top position in LB. Along the way we faced several distinct challenges, each of which shaped a design decision: (i) **a tight compute budget**, addressed by freezing the encoder and exporting heads to ONNX; (ii) **severe class imbalance** ($\approx 499:1$), where rare non-bird taxa—2.1% of the data—carry 30.8% of the score and 28 classes have no training audio at all; (iii) **multiple data-leakage risks** from overlapping audio windows and author concentration; (iv) **overfitting**, which repeatedly punished otherwise reasonable ideas (larger context, embedding mixup, top-1 routing); and (v) **a focal→soundscape domain shift** that makes local validation a very unreliable proxy for the leaderboard. This report documents the implementation, the experiments we ran, the techniques we adopted and discarded, and the lessons that we learned. The final 5-fold ensemble reaches **0.839** private LB.

Contributions. Both authors jointly designed the two-stage pipeline and co-wrote this report. D. Robertson led the data analysis and embedding-space diagnostics—the class-imbalance and leakage audits, the linear-probe and label-free domain classifiers, the Mixture-of-Experts efficiency study, and the domain-bridging experiments (waveform background mixing and iterative pseudo-labeling). J. Obrador led the model training and optimization—the hyperparameter sweeps, the

Focal-Loss and BatchNorm integration, the TTA and 5-fold ensembling pipeline, and the ONNX export and Kaggle deployment.

2 State of the Art

After looking at the implementations and demonstrations of previous winners, it seemed that BirdCLEF competitions converged on three layers. **Foundation models.** Frozen large-scale encoders—BirdNET [1] and Google Perch/Perch-v2 [2]—map waveforms to embeddings already near-linearly separable by species. Perch-v2 is an EfficientNet-family CNN with an internal log-mel front-end. When frozen it trades adaptability for iteration speed and avoids overfitting the encoder to focused audio instead of the usecase of soundscape audio. The strongest BirdCLEF+2025 solutions were won on *preprocessing* rather than architecture. Recurring steps include: 32 kHz mono resampling with **abs-max** waveform normalisation; **segment selection**—choosing *which* seconds of a focal clip to use (the 1st-place solution trained on ~ 20 s chunks “tailored to the target species,” while the top-2% entry found the *middle* 5 s beat highest-RMS or random crops, with random crops for rare classes [7]); **Silero-VAD** [5] speech removal to strip recordist narration (“alien speech”); setting all **secondary labels to 1** for a multilabel target; waveform **MixUp** with element-wise-max labels—the most impactful single augmentation, mimicking overlapping calls—and **background mixing** with real soundscape/ESC-50 [6] noise as an explicit focal→soundscape bridge; **SpecAugment** time/frequency masking; class-imbalance sampling $w_i \propto c_i^\gamma$ ($\gamma \approx -0.5$); per-recording **TopN** post-processing (+1–1.5%). Because our front end is a *frozen* Perch encoder fixed at 5 s/32 kHz, the spectrogram-level knobs do not transfer; but the audio-domain levers—segment selection, MixUp and background mixing *before* embedding, secondary-labels=1, TopN, and pseudo-labeling—do, and we adopt or test several of them (§5). **Heads.** On frozen features, heads range from linear probes to MLPs and Mixture-of-Experts; a sparse top-1 MoE in the Switch-Transformer style [4] promises conditional compute and specialized experts.

3 Methodology I — Data Analysis

We extract frozen Perch-v2 embeddings offline (5 s windows at 32 kHz $\rightarrow$ 1,536-D, mean-pooled per recording) and cache them as `.npy` arrays, sidestepping the Kaggle RAM limit. All downstream analysis and training operate on these vectors.

3.1 Imbalance and the leverage of rare taxa

The label distribution is long-tailed (imbalance ratio $\approx 499:1$; 35,549 recordings; 206 species with audio, 234 scored). Figure 1 shows the decisive asymmetry: Aves is 97.9% of the data but 69.2% of the score, while the non-bird taxa—2.1% of the data—carry 30.8% of the macro score. Of the 28 zero-shot classes (25 insect sonotypes, 3 amphibians), none can be learned from `train.csv`; they remain a permanent ceiling on the achievable score, we attempted to source external data for these missing classes but did not manage in the end.

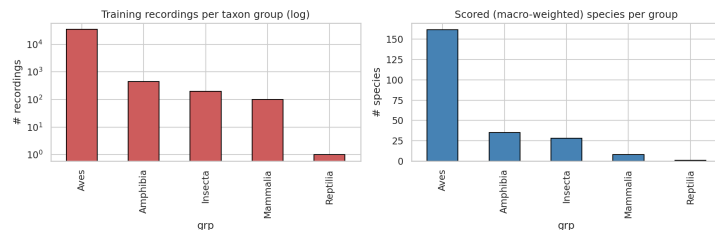


Figure 1: Score leverage by taxon: non-birds are 2.1% of training data but 30.8% of macro-AUC, so robustness on rare, low-resource classes—not bird accuracy—decides the ranking.

3.2 Data leakage and a leakage-safe split

Our second major finding was how easily this dataset leaks. Standard stratified K-fold is invalid here: multiple windows of one recording, or many recordings by one prolific author, can land on both sides of a split and inflate validation AUC. We therefore train with **GroupKFold** ($k=5$) grouped by `filename`, keeping every augmented view of a clip in a single fold; an author-grouped probe confirms the residual leakage is small (a +0.0025 AUC delta).

3.3 Two domain gaps and a saturated representation

A focal→soundscape shift separates clean training clips from the noisy, overlapping test soundscapes, and a second source shift divides curated Xeno-Canto from casual iNaturalist uploads. To gauge their effect, we ran two probes (Figure 2). A bare linear classifier on the frozen embedding already reaches **0.9941** macro-AUC on focal audio (per-group: Aves 0.997, Insecta 1.000, Amphibia 0.985, Mammalia 0.942), i.e. the representation is near-saturated on *in-domain* audio; yet a label-free classifier distinguishes focal from soundscape embeddings at **0.9976** AUC. This explains why our local CV (~ 0.98) sits far above the leaderboard (~ 0.80). A lot of the gap is domain shift, not head capacity. This is a large issue that we faced throughout the project. We’ll return to it when interpreting experiments.

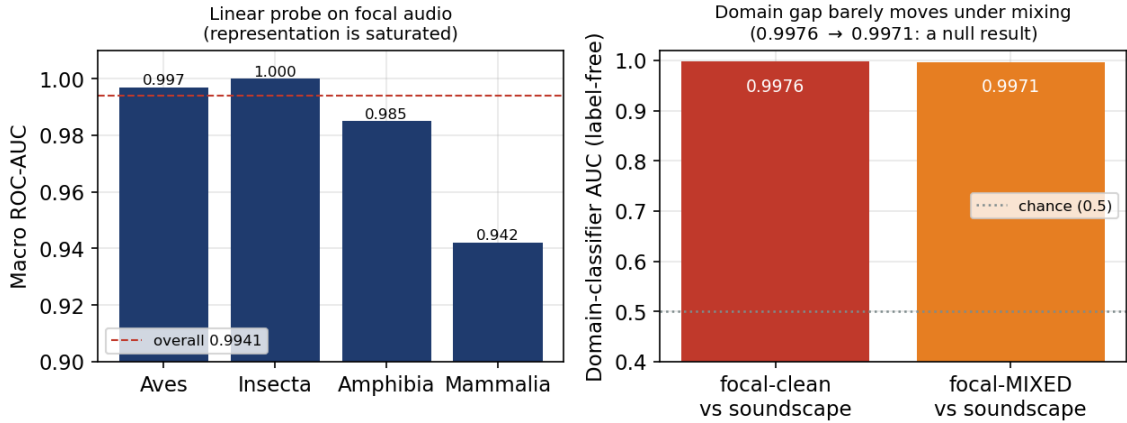


Figure 2: Left: a linear probe nearly saturates focal audio (overall 0.9941). Right: a label-free domain classifier still separates focal from soundscape at 0.9976, and waveform background mixing barely moves it (0.9976→0.9971). Local validation therefore over-states hidden-test performance.

4 Methodology II — Models and Optimization

4.1 Frozen encoder and an efficient runtime

We use frozen 1,536-D Perch-v2 embeddings rather than training spectrogram CNNs end-to-end: the head trains in minutes and the heavy encoder runs once. To meet the 90-minute CPU budget we exported every trained head from PyTorch to a static **ONNX** graph (parity checked to 10^{-5}); the PyTorch runtime was too resource-heavy for the inference window, while ONNX was substantially faster. **Limitation:** the encoder is frozen, so the classifier cannot request finer frequency resolution for a confused species pair.

4.2 From Sparse MoE to Soft Routing

Our initial classification head was a **sparse Top-1 MoE** ($N = 1$): the router selected only the single most confident expert per sample to minimize inference cost and encourage expert specialization. However, this implementation proved **pathological**; the router suffered from “expert

collapse,” where it fixated on a single expert while the others remained dormant. We attempted to cure this early on by introducing a **load-balancing loss**:

$$\mathcal{L}_{\text{lb}} = E \sum_{e=1}^E f_e P_e \quad (1)$$

which successfully prevented collapse and recovered performance parity at **67.6% fewer** multiply-accumulates (Figure 3).

Despite this recovery, we ultimately transitioned to a **soft (dense) MoE** for our final deployment to better handle the multi-label nature of real-world soundscapes. Weighting all experts via a soft routing mechanism allowed for smoother feature blending when multiple species vocalize simultaneously. The final forward pass is defined as:

$$\pi(x) = \text{softmax}(W_{r2} \text{ReLU}(\text{BN}(W_{r1}x))), \quad \hat{y} = \sum_{i=1}^4 \pi_i(x) \text{Expert}_i(x) \quad (2)$$

where each Expert_i is a stabilized MLP. The late-stage integration of **BatchNorm** within the router bottleneck proved essential for stabilizing these continuous gating weights, while a dropout rate of $p = 0.2$ regularized the experts to maximize out-of-domain generalization.

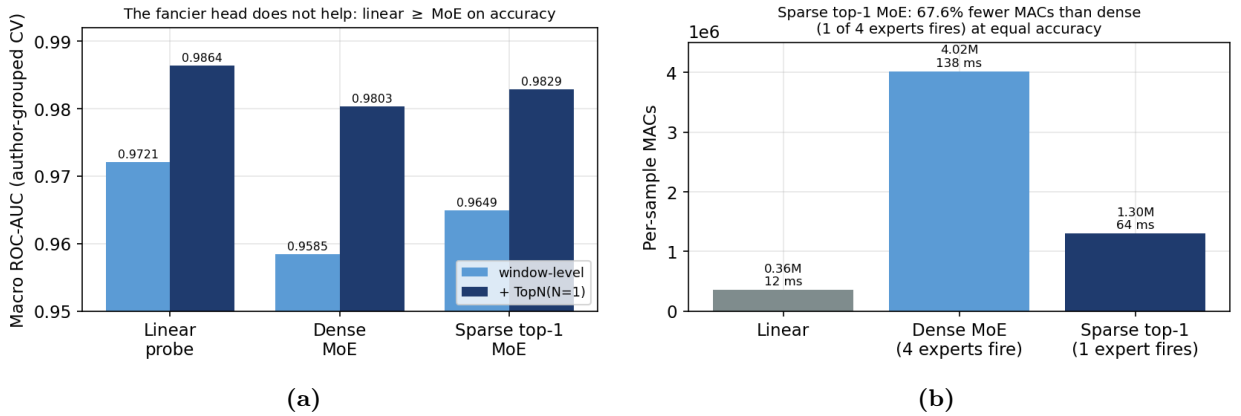


Figure 3: Head comparison on author-grouped CV. (a) Accuracy is statistically flat across linear probe, dense MoE and a load-balanced sparse MoE—the head is not where accuracy comes from. (b) The (fixed) sparse top-1 MoE fires one of four experts, using 67.6% fewer MACs than the dense head.

4.3 Objective: from BCE to Focal Loss

Following a prior winner we first tried binary cross-entropy, but with 233 classes empty for almost every window it was swamped by easy negatives and underperformed. We switched to **Focal Loss** [3], $\mathcal{L}_{\text{focal}} = -\alpha(1 - p_t)^\gamma \log p_t$, which down-weights easy negatives and preserves gradient on rare positives; a proxy-fold sweep (§5) selected $\gamma=1$. The model trains with a softmax + Focal objective; at *inference* we apply a per-class sigmoid (expit), a monotone transform that leaves the per-class macro-AUC ranking unchanged.

4.4 Inference pipeline

At test time Perch embeddings are computed on 50% overlapping 2.5s-hop windows; probabilities are averaged across windows and the five ONNX folds, then sharpened per-recording with TopN(N=1). A geographic mask down-weighting species outside the Pantanal box was tested but *not* used in the final submission as it tanked performance.

5 Experiments

Table 1 summarizes every major technique we tried, its outcome, and why. The remainder of this section details the most informative experiments. All sweeps used TTA-unwrapped data, Focal Loss, the MoE, 5 proxy epochs and fold 0 as validation.

Table 1: Techniques attempted during the competition, with outcomes and rationale.

Technique	Outcome	Why
PyTorch $\rightarrow$ ONNX runtime	Adopted	PyTorch too heavy for the 90 min CPU window; ONNX much faster.
Binary cross-entropy (winner’s choice)	Rejected	Drowned by easy background negatives under heavy imbalance.
Focal Loss ($\gamma=1$)	Adopted	Focuses gradient on rare, faint positives; best LB.
50% overlapping TTA	Adopted (biggest gain)	Rescues clipped transients; +0.025 public AUC, despite a winner discouraging it.
5-fold ensembling	Adopted	Cancels fold-specific blind spots: 0.742 $\rightarrow$ 0.839 private.
BatchNorm in router (added late)	Adopted	Stabilized routing; improved LB after re-tuning.
Embedding mixup (70/30, Beta)	Rejected	Over-regularized into “alien” embeddings; -0.034 LB.
Background mixing (wave-form)	Rejected	Naive overlay; null on the domain metric (“alien” sounds).
Larger context (ctx 1 $\rightarrow$ 5)	Rejected	Overfit macro-environmental site acoustics; worse LB.
Top-1 sparse MoE ($N=1$)	Rejected	Router collapsed onto one expert (fixed only later, with load balancing).
Geographic mask	Rejected	No leaderboard improvement.
Pseudo-labeling (Noisy Student)	Explored, dropped	Marginal gains and too complex/brittle to justify.

5.1 Augmentation: TTA wins, mixup fails

We compared (A) a CE baseline, (B) embedding mixup (Beta(0.4,0.4), 70/30), and (C) 50%-overlap TTA with Focal training (Figures 4–5). **TTA was our single biggest improvement** in local loss and gained +0.025 public AUC; **mixup lost** -0.034 . The mixup failure is instructive: it interpolates *already compressed* Perch coordinates $\tilde{x}=\lambda x_i+(1-\lambda)x_j$, producing “alien bird” vectors matching no real mixture of calls, so the model under-learns real recordings despite the *highest* embedding CV (0.9994). TTA, by contrast, perturbs the audio itself: overlapping crops expose multiple phase alignments of a call, enforcing translation invariance that survives to the test set.

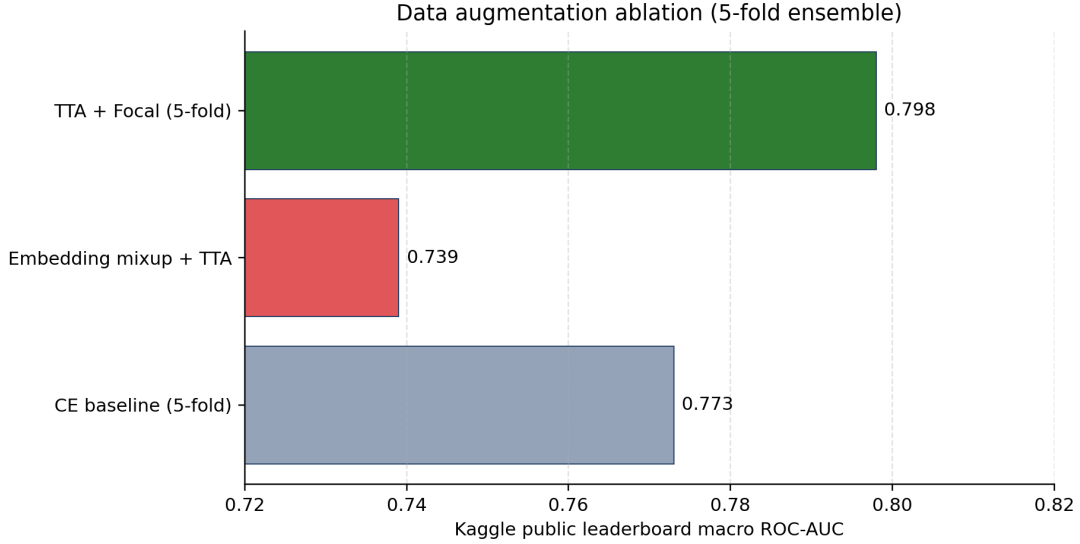


Figure 4: Augmentation ablation on the Kaggle public LB (5-fold ensemble). A: CE baseline; B: embedding mixup; C: TTA + Focal training.

Model	Variant	Headline score	Δ vs baseline	Rank
A	CE baseline (no augmentation)	0.773	+0.000	2
B	Embedding mixup + TTA	0.739	-0.034	3
C	TTA windows + Focal Loss	0.798	+0.025	★ 1

Variant	F0	F1	F2	F3	F4	Mean	Std	Rank
A — CE baseline	0.9926	0.9991	0.9995	0.9996	0.9998	0.9981	0.0028	2
B — Mixup + TTA	0.9986	0.9996	0.9995	0.9997	0.9997	0.9994	0.0004	★ 1
C — TTA + Focal	0.9788	0.9834	0.9857	0.9874	0.9895	0.9850	0.0037	3

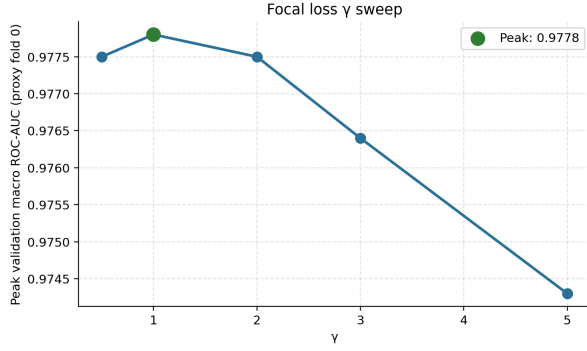
Figure 5: Detailed augmentation ablation. Top: headline Kaggle scores. Bottom: embedding-level 5-fold CV. Mixup tops CV but bottoms the LB—an inversion driven by the domain gap, since only TTA modifies the data.

5.2 Hyperparameter sweeps and the context trap

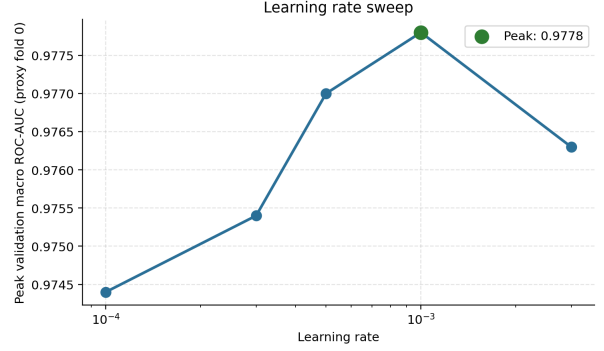
Proxy-fold sweeps over Focal γ , learning rate and temporal context (Figures 6–8b) maximize single-epoch peak validation AUC and, being in-domain, do not always transfer. The clearest example is **temporal context**: expanding the window (ctx 1 $\rightarrow$ 5; 1,536 $\rightarrow$ 7,680-D) won proxy validation but *crashed* the leaderboard (0.783 vs. 0.839), because the wider window let the lightweight head memorize site-level background acoustics (wind, rain) instead of biological transients. We kept ctx 1 as a deliberate bottleneck.

Hyperparameter	Best setting	Peak Val AUC
Focal loss γ	1.0	0.9778
Weight decay	1e-5	0.9781
Learning rate	1e-3	0.9778
Context window	5 windows	0.9899

Figure 6: Proxy-fold sweep winners (single-epoch peak validation AUC); these do not always transfer.

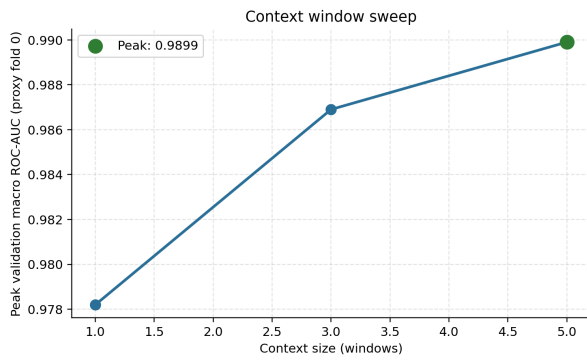


(a) Focal γ

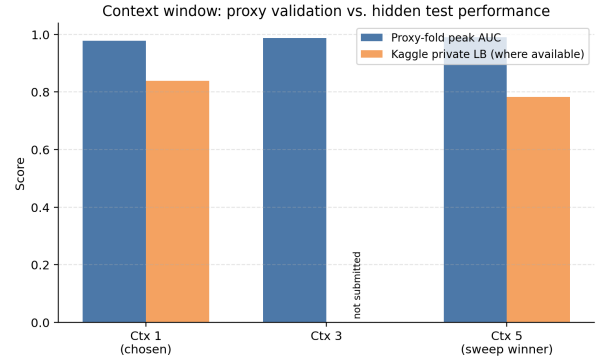


(b) Learning rate

Figure 7: Hyperparameter sweeps on proxy fold 0.



(a) Context window

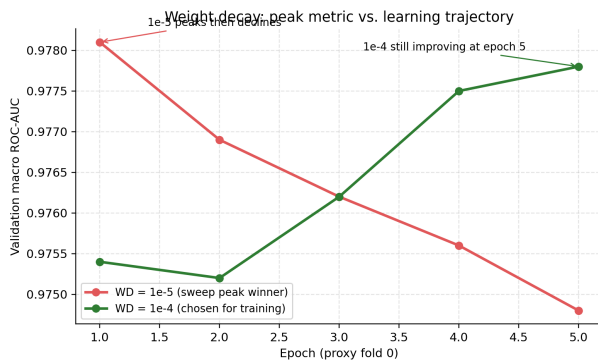


(b) Proxy vs. private LB

Figure 8: Temporal-context ablation: larger context wins proxy validation but loses the hidden test.

5.3 Weight decay and overfitting trajectories

The weight-decay sweep nominally favors $\lambda=10^{-5}$ (peak Val AUC 0.9781 at epoch 1), but per-epoch analysis shows early overfitting—Val AUC declines every epoch while train loss falls (Figures 9a–9b). We deployed $\lambda=10^{-4}$, which improves monotonically, choosing on the *trajectory* rather than the single-epoch peak. The same saturation signature appears in the pooled learning curves (Figures 10a–10b): the head fits focal audio within a few epochs and further training is mild overfitting. **Caveat:** these curves monitor Focal Loss on precomputed embeddings, not soundscape performance.



(a) WD trajectories

Epoch	1e-5 Val AUC	1e-4 Val AUC	Δ (1e-4 - 1e-5)	1e-5 Val loss	1e-4 Val loss	Note
1	0.9781	0.9754	-0.0027	1.3018	1.3341	1e-5 peaks (early best)
2	0.9769	0.9752	-0.0017	1.4273	1.3392	1e-5 Val AUC declining
3	0.9762	0.9762	+0.0000	1.5125	1.3529	Curves converge
4	0.9756	0.9775	+0.0019	1.5651	1.3659	1e-4 overtaking
5	0.9748	0.9778	+0.0030	1.6090	1.3329	1e-4 peaks; ahead at end

(b) Per-epoch Δ

Figure 9: Weight decay 10^{-5} vs. 10^{-4} on proxy fold 0: 10^{-5} peaks early then overfits; 10^{-4} improves monotonically.

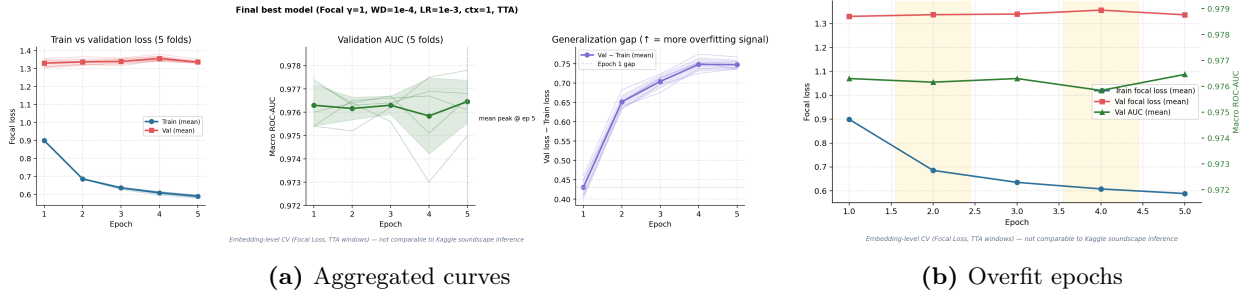


Figure 10: Aggregated 5-fold learning curves (final model); bands mark epochs where train loss improved but mean Val AUC did not.

5.4 Domain-bridging attempts: background mixing and pseudo-labeling

Two ideas aimed squarely at the domain gap, and both disappointed. **Background mixing** overlays a random soundscape onto each focal clip on the waveform ($\text{SNR} \in [3, 20]$ dB) before embedding; the domain classifier barely moved ($0.9976 \rightarrow 0.9971$, Figure 2), a **null**. We attribute this to a naive overlay that creates the same “alien” artifacts as embedding mixup; a better scheme would mix only no-call/ambient soundscape segments (or do scene-aware mixing) so the blended clip stays acoustically plausible—left as future work. **Iterative pseudo-labeling** (Noisy-Student style) labels unlabeled soundscapes, keeps high-confidence chunks, and retrains; held-out target-domain confidence did rise across iterations (Figure 11), but the gains were marginal and the loop was complex and brittle, so we **did not adopt it** in the final system.

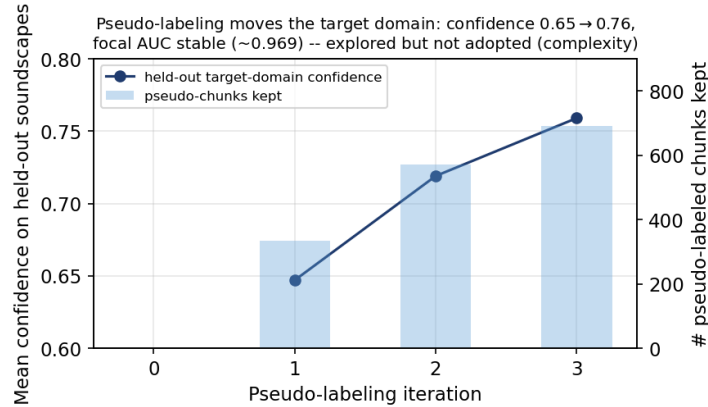


Figure 11: Iterative pseudo-labeling (explored, not adopted): held-out target-domain confidence rises $0.65 \rightarrow 0.76$ across three iterations, but the practical benefit did not justify the complexity.

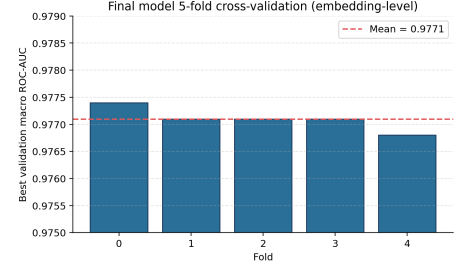
6 Results

6.1 Final configuration and cross-validation

The deployed model (Figure 12a) trains five fold heads for 5 epochs: Focal Loss ($\gamma=1$), Adam ($\text{lr} = 10^{-3}$, $\text{wd} = 10^{-4}$), context 1, TTA-unwrapped samples, batch 64. Embedding-level CV: **0.9771** $\pm$ **0.0005** macro-AUC (Figure 12b); each fold exports an ONNX graph and the five are averaged uniformly.

Component / Parameter	Final Checked Specification	Engineering Purpose
Feature Extraction	Google Perch-v2 (frozen)	Linear probing baseline to prevent microphone overfitting.
Temporal Context	Context 1 (1,536 dimensions)	Restricts optimization to immediate acoustic transients; avoids long-term noise shift.
Augmentation (Train)	Symmetric 50% Overlapping TTA	Enforces temporal translation invariance across window boundaries.
Routing Topology	2-Layer Bottleneck (1536→256→4)	Filters dimension noise; maps non-linear routing boundaries.
Regularization	BatchNorm1d + Dropout(0.2)	Prevents expert routing monopoly and stabilizes internal distributions.
Weight Decay	1e-4 (Adam L2)	Chosen over 1e-5 from sweep: progressive Val AUC improvement vs early peak-then-decline (proxy fold 0).
Optimization	Adam (LR=10 ⁻³ , WD=1e-4) + Focal Loss ($\gamma=1.0$)	Per-parameter rate adaptation; dynamic easy-negative suppression.
Deployment Mode	5-Fold Out-of-Fold Ensemble	Uniform probability averaging to minimize fold-specific variance.

(a) Final configuration



(b) Per-fold Val AUC

Figure 12: Deployed model specification and its 5-fold cross-validation (mean 0.9771).

6.2 Leaderboard progression

Table 2 and Figure 13 trace the submission history. Ensembling was a reliable win (single fold 0.742 $\rightarrow$ ensemble 0.839 private); Focal Loss, TTA training and the late BatchNorm each moved the needle. A recurring lesson is the unreliability of any single in-domain signal: the best-*private* model (0.839) was only middling on the *public* board (0.798), where the no-BatchNorm run (0.806) and Focal $\gamma=2$ (0.805) scored higher—so we relied on ensembling and conservative choices rather than chasing the public number. On the private leaderboard, our final score of 0.839 placed us rank 3069 / 4084 teams.

Table 2: Kaggle submission timeline (macro ROC-AUC), earliest to final. Private confirmed post-close. Δ Priv. is each run’s private score minus the single-fold baseline (0.742).

Experiment	Configuration	Priv.	Pub.	Δ Priv.
Single-fold MoE	CE; lr10 ⁻³ , wd10 ⁻⁴ ; 10 ep; 1 fold	0.742	0.739	— (ref)
5-fold ensemble	Same CE baseline; 5-fold OOF mean	0.761	0.773	+0.019
Mixup + TTA	Embedding mixup + TTA windows	0.770	0.766	+0.028
+ Focal ($\gamma=2$)	CE $\rightarrow$ Focal $\gamma=2$	0.792	0.805	+0.050
MoE + TTA + Focal	TTA training + Focal; 10 ep	0.819	0.798	+0.077
Large context	ctx 5; Focal $\gamma=0.5$; wd10 ⁻⁵ ; lr10 ⁻⁴	0.783	0.780	+0.041
MoE w/o Batch-Norm	Focal $\gamma=0.5$; wd10 ⁻⁵ ; lr10 ⁻⁴ ; ctx 1	0.835	0.806	+0.093
Final model	Focal $\gamma=1$; wd10 ⁻⁴ ; lr10 ⁻³ ; ctx 1; BN; 5 ep	0.839	0.798	+0.097

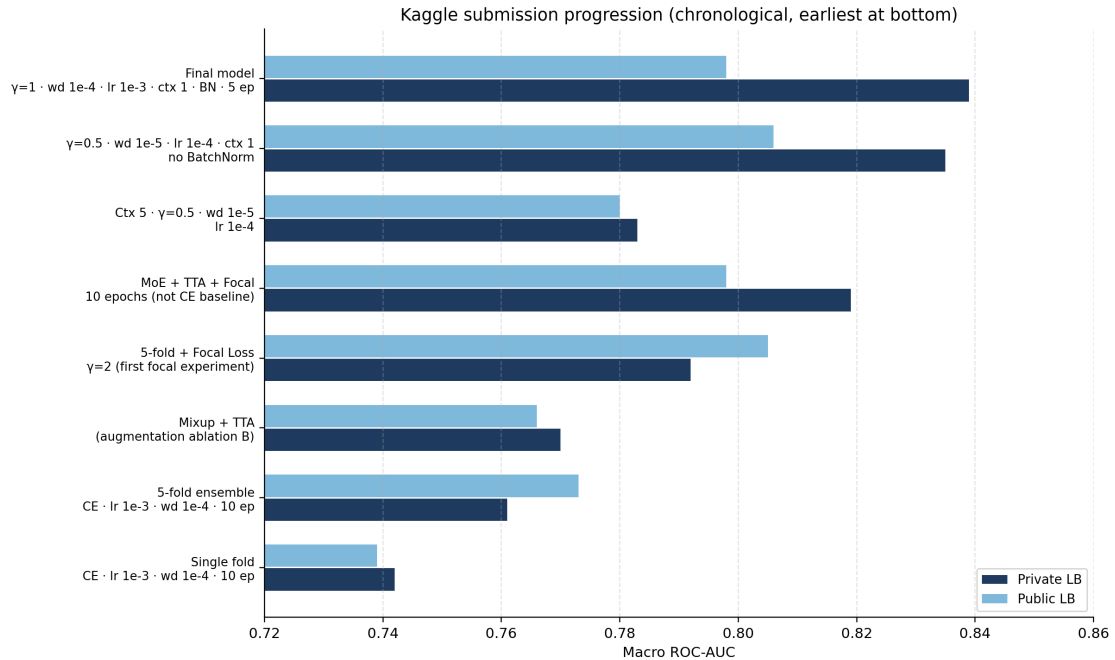


Figure 13: Submission progression (chronological), labelled with key hyperparameters per run.

7 Discussion

A few themes connect our results. **What worked were data-side and variance-reducing moves:** TTA (perturbing the audio), Focal Loss (re-weighting an imbalanced objective), Batch-Norm (stabilizing routing), and ensembling. **What failed shared a cause**—they either created unrealistic training signal (embedding mixup, naive background mixing: “alien” samples) or gave the lightweight head more room to overfit (larger context, unbalanced top-1 routing). The data analysis ties several failures together: with a near-saturated focal representation, augmentations that merely re-mix embeddings cannot help, and the focal→soundscape domain shift is why local CV repeatedly mis-ranked models relative to the leaderboard. We treated the public LB cautiously for the same reason.

The zero-shot ceiling. Twenty-eight of the 234 scored classes (25 insect sonotypes and 3 amphibians) have no training audio at all, so a purely supervised head can only rank them at chance. Because macro-AUC weights every class equally, these classes impose a hard ceiling that is independent of head quality: if the 206 learnable classes were scored at our in-domain level (≈ 0.95) while the 28 unreachable classes sat at chance (0.5), the macro average would be capped near $(206 \times 0.95 + 28 \times 0.5)/234 \approx 0.90$ —and substantially lower once the focal→soundscape gap is added back. These untrained classes fall entirely within the non-bird taxa that, despite being 2.1% of the data, carry 30.8% of the score, so they are precisely the high-leverage classes we most need and least can fit. This is why we attempted (unsuccessfully) to source external recordings for them, and why a class-conditional zero-shot route—exploiting Perch’s own pretrained vocabulary, or training separate insect/amphibian models as the 2025 winner did—is, in our view, the single most promising direction for breaking past ~ 0.84 .

Limitations. We cannot directly measure soundscape accuracy (those labels are the hidden test), so domain-gap evidence is indirect (focal saturation plus a label-free separation probe). The 28 zero-shot classes are unreachable by any supervised head. Background mixing was a null under our naive implementation, and pseudo-labeling, though promising in principle, was too complex to land within the project budget.

8 Conclusions

We built a frozen-Perch + soft-MoE pipeline, exported it to ONNX for offline inference, and optimized it to **0.839** private macro-AUC. The improvements that mattered were overlapping-window TTA (our largest gain), Focal Loss over a binary cross-entropy baseline, a late BatchNorm stabilization, conservative temporal context, and 5-fold ensembling. Equally informative were the failures—embedding mixup, waveform background mixing, larger context, top-1 routing collapse, geographic masking, and an abandoned pseudo-labeling loop—most of which trace back to overfitting a saturated representation or to the focal→soundscape domain shift that decouples validation from the leaderboard. **Future work** follows directly: a scene-aware background-mixing scheme that avoids “alien” blends; a simpler, better-validated pseudo-labeling loop; load-balanced sparse routing for cheaper inference; and learned ensemble weighting or temporal smoothing on long soundscapes.

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